

SAE INDIA

SOUTHERN SECTION

Bicycle Design Challenge 2025-26



Design Report - 3



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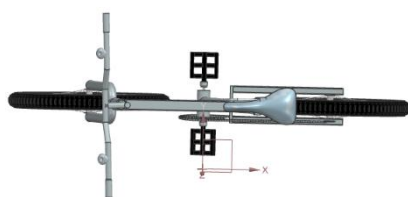
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OVERVIEW

Our Z – frame bicycle Address

The bicycle features a unique Z-frame with a minimalist structure, including front and rear wheels of equal size, a standard chain drive mechanism, and a crank set with pedals. The saddle is mounted on an inclined seat post, and the handlebars are angled for ergonomic use. A local coordinate system near the crank indicates the model used for kinematic or structural analysis.



IDENTIFY TOOLING REQUIREMENT

◇ **Frame Manufacturing Tool:**

- **Tube Bending Machine:**
Helps bend the mild steel pipes into the correct shape without cracking.
 - **Laser or Plasma Cutter:**
Cuts the mild steel parts with high accuracy into the shapes we need.
 - **Holding Jigs:**
Special holders keep the frame parts in the right position during welding.
 - **Welding Machine:**
Joins the mild steel frame parts together strongly and neatly.
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◇ **Assembling the Bicycle:**

- **Torque Wrench:**
Tightens bolts to the right level to loose or to tight can cause problems.
 - **Bearing Press Tool:**
Helps fit parts like the pedal crank or steering bearings smoothly into the frame.
 - **Chain Tool:**
Used to connect and adjust the bike chain properly.
 - **Pedal and Crank Tools:**
Attach or remove pedals and cranks easily and safely.
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◇ **Fitting the Seat and Suspension:**

- **SpringTool:**
For installing the spring under the seat, making the ride smoother.
 - **Seat Clamp Tool:**
Allows you to adjust the seat height for different rider sizes.
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◇ **Setting Up Wheels:**

- **Wheel Stand:**
Makes sure the wheels spin straight and are balanced before installing.
 - **Tire Tools & Pump:**
Help fix the tires onto the wheel and fill them with air.
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◇ **Final Fit for Comfort:**

- **Handlebar Adjuster:**
Helps align the handlebar so it's comfortable for the rider.
 - **Grip Fitting Tool:**
Slides the rubber handle grips on securely.
 - **Kickstand Tool:**
Helps attach the kickstand on the side of the mild steel frame properly.
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◇ **Testing and Measurement Tools:**

- **Callipers / Micrometre:** For precise measurement of parts.
 - **Weighing Scale:** To measure weight to ensure the bicycle stays lightweight.
 - **Jig/Alignment Gauge:** To ensure the alignment of the frame and wheels.
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◇ **Basic Tools Needed:**

- Small and medium Phillips screwdrivers.
- Combination wrench set including mm from 7, 8, 9, 10, 11, 12, 13, 14, 15, 16
- to 17mm wrenches
- Regular pliers.
- A Needle-nose pliers
- Req. Small and medium Vise-Grip pliers
- Plumbers' (also called "water pump") pliers
- Diagonal cutters should be used
- 6-inch adjustable wrench
- 12-inch adjustable wrench
- A Ballpeen hammer (8-ounce size is about right)
- Plastic or rubber mallet
- Req. Hacksaw (photo, right)
- Course and fine file (flat and half-round)
- Metal punches and chisels
- Outside/inside the calliper
- Awl
- Knife
- Scissors
- 8, 9, 10mm "Y" tool
- Allen wrench set comprised of 2, 3, 4, 5, 6, 7, 8 and 10mm
- wrenches
- 4, 5, 6mm combination Allen wrench (photo, right) or folding
- Allen wrench
- set
- Metric/English tape measure
- Floor pump with gauge
- Chain tool (compatible with your chain)
- Spoke wrench (to fit your spoke nipples)
- 3 tire levers

- Pedal wrench (photo, middle) 2 each: 13, 14, 15, 16, 17, 18mm cone
- wrenches
- Axle vise

IDENTIFY GAUGE REQUIREMENT

- **Frame Alignment Gauge:**

Used to check that the Z-frame structure is straight and properly aligned during assembly, ensuring a smooth and stable ride.

- **Spoke Tension Gauge:**

Helps measure and adjust the tension in the wheel spokes, especially important for 360-degree wheels to stay balanced and strong.

- **Torque Gauge / Torque Wrench:**

Used to tighten bolts and screws (like on the handlebar, seat, and pedals) to the exact level, so they're secure but not over-tightened.

- **Brake Rotor Alignment Gauge:**

Used to check that the V-brake pads are aligned properly with the rim, ensuring smooth and safe braking.

- **Tire Pressure Gauge:**

Measures the air pressure inside the tires, helping you set the right pressure for a smooth and safe ride.

- **Suspension Travel Gauge:**

Used to set and check the movement of the seat-mounted spring suspension, ensuring it works correctly and absorbs bumps well.

RAW MATERIAL USED FOR PROTO SAMPLES

- **Steel Tubing**

We used strong steel pipes to build the main frame of the bicycle. Steel is affordable, easy to work with, and strong enough for testing.

- **Plastic & Polymer Parts**

Materials like plastic were used to make parts such as the pedals, seat cover, and handle grips. They are light, easy to shape, and suitable for quick prototype builds.

- **Rubber Components**

Rubber was used in the tires and hand grips. It gives good grip and absorbs shocks for a smoother ride.

- **Aluminium Lugs (Connectors)**

Some joints or connectors in the frame may be made from aluminium, which is light and resists rust.

- **Foam or Soft Padding**

Soft foam was added to the seat and handlebar areas to make the ride more comfortable during testing.

- **Steel Forks**

The front fork (which holds the front wheel) is also made of steel to handle loads and bumps well.

- **Bolts, Nuts, and Fasteners**

All the screws, nuts, and bolts used in the assembly are made from metal (mostly steel) to keep parts tightly fixed.

PROTOTYPE PREPARATION

- **Design Finalization:**

A full digital model (CAD drawing) of the bicycle was made, including all parts.

- **Material Selection:**

The main frame is made using mild steel for strength. Wheels are made of nylon or aluminium to keep them lightweight and durable.

- **Manufacturing:**

The frame parts were made using welding. Small parts (like mounts or clips) were created using 3D printing if required.

- **Assembly:**

All the bicycle parts like handlebars, pedals, V-brake system, and suspension were carefully assembled using proper tools. Special care was taken to check that all movements (pedalling, steering, braking) are smooth and functional.

PROTOTYPE VALIDATION

- **Structural Testing:**

The frame strength was checked to make sure it can carry weight and handle daily use.

The alignment of wheels and frame was tested for balance.

- **Functional Testing:**

A test ride was done to check how well the bicycle handles turns, brakes, and how smooth the suspension.

- **Durability Testing:**

The bicycle was tested under stress and repeated use to make sure it lasts over time.

- **User Feedback:**

Different users were asked to ride and give feedback on comfort, ease of use, and design.

Based on their comments, some improvements were made.

- **Final Validation:**

Before approving the prototype, the bicycle was checked again for safety, performance, and design quality to ensure everything meets the required standard

CONCEPTUAL SKETCHES



MANUFACTURING PROCESSES USED

◇ Material Preparation

- **CNC Cutting:**
Used to accurately cut steel tubing for the frame and fork parts.
 - **Laser Cutting:**
Applied for cutting complex frame joints or mounting points (like for the pedals and suspension connections).
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◇ Frame Fabrication

- **Gas Welding:**
Used to join the steel tubes of the bicycle frame securely. This ensures high strength and clean welds.
 - **Tube Bending:**
Helps form the angled and curved sections of the frame to achieve the unique ergonomic and structural design.
 - **Injection moulding:**
Used to make plastic parts like pedals, handle grips, and seat parts with precision and consistency.
 - **3D Printing:**
Useful for prototyping or producing small custom parts, such as folding joints, cable clips, or design covers.
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◇ Component Manufacturing

- **CNC Machining:**
Used for high-precision parts such as the bottom bracket housing, gear teeth, and crank set parts.

- **Forging:**
Applied to produce strong parts like crank arms or pedal bodies, which must withstand repeated force.
 - **Casting:**
Used to make hub bodies and gear carriers, especially for compact or detailed parts.
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◇ **Surface Finishing**

- **Powder Coating:**
The frame and metal parts are coated for a durable and scratch-resistant finish in desired colours.
 - **Polishing:**
Applied to steel parts like the handlebar or forks for a clean, shiny surface.
 - **Anodizing:**
Improves corrosion resistance and gives a stylish appearance to aluminium components.
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◇ **Assembly**

- **Mechanical Assembly:**
The bicycle is assembled by mounting the wheels, handlebars, seat, pedals, and chain drive using proper tools.
 - **Brake System:**
A standard V-brake system is installed for effective and low-maintenance braking performance.
 - **Suspension Fitting:**
If present, locking suspension components are installed and tested for smooth operation.
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◇ **Testing & Quality Control**

- **Functional Testing:**
The fully assembled bicycle is tested for alignment, smooth folding, brake efficiency, and ride comfort.
 - **Durability Testing:**
Frame and components undergo stress and load testing to ensure long-term performance.
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◇ Final Inspection and Packaging

- **Quality Inspection:**
Final checks ensure the bike is safe, defect-free, and meets design standards.
- **Protective Packaging:**
The finished bicycle is securely packaged using foam, wraps, and cardboard to protect during shipping.

MODIFICATION BASED ON PROTOTYPE TESTING/ DFA/DFM



PROTOTYPE

Changes After Prototype Testing

Stronger Frame:

- During tests, the centre part of the frame bent a little under pressure.
- So, we added extra support bars to make it stronger and better at carrying weight.

Better Seat Post:

- The first seat could not be easily adjusted for Length.
- We replaced it with a quick-release seat post, so riders can adjust the height quickly and comfortably.

Improved Pedal Grip:

- The original plastic pedals were too slippery.
- We replaced them with rubber or textured pedals for a better grip and safer riding.

Design for Assembly - DFA

Modular Design:

- The bicycle is now built using main sections like the frame, fork, and rear part.
- This allows for faster assembly and easier repair or replacement of parts.

Standardized Fasteners:

- Instead of using many different bolts and screws, we now use just 2 or 3 common types.
 - This means less tool-changing, making the job easier and faster.
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Design for Manufacturing - DFM

Simpler Tube Shapes:

- We reduced the number of complex bends in the frame tubes.
- Now we use straight or single-bend tubes, which are easier and faster to cut and weld.

Improved Welding Process:

- We rearranged the places where welding is done, so that fewer welds are needed and they are easier to do.
- This helps save time and keeps quality more consistent.

3D Printed Samples:

- For tricky parts like folding joints, we first made them using a 3D printer.
- This allowed us to test and fix the design before making expensive metal Molds for production.

DESIGN REPORT REVIEW & INSPECTION REPORTS FOR BICYCLE / SYSTEM / PART

◇ Project Overview:

The Z-frame bicycle is a modern and stylish city bike designed for better stability and visual uniqueness. Unlike traditional bike shapes, this bicycle uses a Z-shaped frame that offers both strength and space efficiency.

Design Goals:

- **Comfortable to Ride** in urban roads
- **Easy to maintain** and assemble
- **Sustainable Materials** with minimal waste

- **Simple Aesthetic** with modern appeal
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◇ **Design Specifications**

Frame Design:

- **Shape:** Z-shaped for structural rigidity
- **Material:** Mild steel
- **Type:** Rigid frame

Components:

- **Wheels:** 26-inch standard road tires
- **Brakes:** V-brakes, easy to maintain and affordable
- **Handlebars:** Straight type for upright posture
- **Seat:** Adjustable, ergonomic cushion
- **Pedals:** Non-slip plastic or rubber

Weight and Size:

- **Total Weight:** Approx. 10–14 kg
 - **Dimensions:** Similar to standard bicycles, fits easily in apartments or on public transport
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◇ **Design Process**

Concept Development:

- Started with basic sketches and tried different frame shapes
- Chose the Z-frame for its unique look and strength
- Created 3D CAD models for simulation

Simulation & Testing:

- Used FEA (Finite Element Analysis) to check the frame's strength
- Simulated riding conditions like bumps and rider load

- Made a prototype and did real-world riding tests

◇ Modifications

Prototype Testing Feedback: Summarize feedback gathered during prototype testing.

Implemented Changes: List the changes made based on testing results, focusing on performance, usability, and manufacturability.

◇ Design Validation

Testing Results: Provide details of the testing conducted (ride tests, stress tests, etc.) and their outcomes.

User Feedback: Summarize insights from users who tested the bicycle, focusing on comfort and functionality

◇ Inspection Checklist

Visual Inspection:

- Frame: No visible cracks or damage
- Welds: Clean and strong
- Paint: Even coating, no scratches

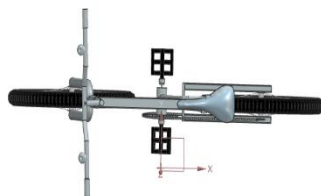
Functional Testing:

- Brakes: Respond quickly and stop smoothly
- Suspension: Not included in this model (rigid frame)

◇ Detailed Inspection Results

Component	Condition	Notes
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Frame	Excellent	Clean welds, strong structure, visually clean
Wheels	Good	Spinning freely, properly aligned
Brakes	Good	Required minor tightening for optimal performance
Handlebars/Seat	Good	Seat is soft and secure; handlebar grip is firm
Pedals/Crank set	Good	Smooth rotation, no looseness



CONCLUSION

This report ensures the Z-frame bicycle has been carefully designed, tested, and inspected to meet quality and usability standards. It is now ready for users, offering a safe and stylish ride for city environments.

thank you 